

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398452
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Chen; Yun-Heng et al.

Metal mask

Abstract

A method of manufacturing a metal mask includes calendering a metal material, so as to form a metal mask substrate, where the metal mask substrate includes a surface and a plurality of grooves formed in the surface, and the grooves all extend in a direction. The surface has at least one sampling region, while at least two grooves are distributed in the sampling region, where an average area ratio of the area of the grooves within the sampling region to the area of the sampling region ranges between 45% and 68%.

Inventors: Chen; Yun-Heng (Hsinchu County, TW), Lin; Wen-Yi (Hsinchu County, TW)

Applicant: Darwin Precisions Corporation (Hsinchu County, TW)

Family ID: 1000008782480

Assignee: DARWIN PRECISIONS CORPORATION (Hsinchu County, TW)

Appl. No.: 17/964176

Filed: October 12, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230037092 A1	Feb. 02, 2023

Foreign Application Priority Data

TW	110121496	Jun. 11, 2021
----	-----------	---------------

Related U.S. Application Data

division parent-doc US 17577046 20220117 US 11685985 child-doc US 17964176

Publication Classification

Int. Cl.: C23C14/04 (20060101); B05C21/00 (20060101); B21B1/22 (20060101); B21H7/00 (20060101); B21H8/00 (20060101); B23P15/00 (20060101); C23C14/24 (20060101); G03F7/00 (20060101)

U.S. Cl.:

CPC C23C14/042 (20130101); B05C21/005 (20130101); B21B1/227 (20130101); B21H7/00 (20130101); B21H8/005 (20130101); B23P15/00 (20130101); C23C14/24 (20130101); G03F7/0015 (20130101); Y10T29/49826 (20150115); Y10T29/49863 (20150115)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5105638	12/1991	Mito et al.	N/A	N/A
8656859	12/2013	Hong	118/504	H10K 71/166
8746169	12/2013	Lee	118/504	H10K 71/166
8881676	12/2013	Hong	118/504	C23C 14/12
9192959	12/2014	Hong	N/A	C23F 1/02
9583708	12/2016	Ko	N/A	C23C 14/042
9666837	12/2016	Lee	N/A	H10K 71/00
9711724	12/2016	Baek	N/A	C23C 14/042
9975134	12/2017	Ko	N/A	H01J 11/48
10014471	12/2017	Baek	N/A	C23F 1/02
10084133	12/2017	Chen	N/A	B05C 21/005
10439170	12/2018	Lin	N/A	C23C 14/042
11578400	12/2022	Zhang	N/A	H10K 71/166
11732347	12/2022	Okamoto	118/712	C23C 14/042
2014/0113144	12/2013	Loth et al.	N/A	N/A
2015/0034005	12/2014	Ko	118/504	C23C 14/042
2018/0065162	12/2017	Mikami et al.	N/A	N/A
2018/0186161	12/2017	Morikawa et al.	N/A	N/A
2019/0337031	12/2018	Jeon et al.	N/A	N/A
2021/0355586	12/2020	Mikami et al.	N/A	N/A
2022/0325397	12/2021	Miyatani	N/A	C23C 14/042

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108025581	12/2017	CN	N/A
201840881	12/2017	TW	N/A

OTHER PUBLICATIONS

Yang et al., “Metal Mask and Inspecting Method Thereof”, U.S. Appl. No. 17/523,947, filed Nov. 11, 2021. cited by applicant

Lin et al., “Method of Preparing Metal Mask Substrate”, U.S. Appl. No. 17/524,076, filed Nov. 11, 2021. cited by applicant

Primary Examiner: Pence; Jethro M.

Attorney, Agent or Firm: WPAT, PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a Divisional Application of the U.S. application Ser. No. 17/577,046, filed Jan. 17, 2022, which claims priority to Taiwan Application Serial Number 110121496, filed Jun. 11, 2021, all of which are herein incorporated by reference in their entireties.

BACKGROUND

Technical Field

(1) The present disclosure relates to a processing tool configured for patterning and a manufacturing method thereof. More particularly, the present disclosure relates to a metal mask and a manufacturing method thereof.

Description of Related Art

(2) Nowadays, some display panels are manufactured by evaporation and using a metal mask, in which the metal mask has a plurality of openings. During the evaporation, plumes can pass through the openings and be deposited on a target substrate (e.g., a glass plate), so as to form a patterned layer. A conventional metal mask is usually manufactured by photolithography, so that the openings are defined by a photoresist pattern after developing.

(3) Current display panels have been developed towards a high resolution trend. Accordingly, the pitches among the openings of the metal mask need to be reduced, so that the metal mask can be used for manufacturing a high resolution display. However, once the pitches among the openings are reduced, the part of the photoresist pattern covering the area between two adjacent openings will also be reduced, so that the photoresist pattern may not completely cover the area between two adjacent openings during etching, resulting in the subsequent formation of some large and ineligible openings, even multiple openings communicating to each other, thereby causing a finished metal mask that is not suitable for or not capable of manufacturing the high resolution display panel.

SUMMARY

(4) At least one embodiment of the disclosure provides a metal mask, which can be used for manufacturing a display panel.

(5) A metal mask according to at least one embodiment of the disclosure includes a surface, a plurality of grooves and a plurality of openings. The surface has at least one sampling region. The grooves and the openings are formed in the surface, and the grooves all extend in a direction, where an average area ratio of the area of the grooves within the sampling region to the area of the sampling region ranges between 45% and 68%.

(6) When the average area ratio ranges between 45% and 68%, the surface of the metal mask substrate can have suitable roughness, in which the suitable roughness facilitates a qualified metal mask having eligible openings, so that the metal mask can be used for manufacturing the display panel.

(7) It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
- (2) FIG. 1A is a schematic view of a calendering process employed by a method of manufacturing a metal mask according to at least one embodiment of this disclosure.
- (3) FIG. 1B is a schematic partial enlarged view of the surface of the calendering roller in FIG. 1A.
- (4) FIG. 2A is a schematic top view of the metal mask substrate formed via the calendering process in FIG. 1A.
- (5) FIG. 2B is a schematic top view within one of the sampling regions of the metal mask substrate in FIG. 2A.
- (6) FIGS. 3A and 3B are schematic cross-sectional views of the metal mask substrate in FIG. 2A during photolithography.
- (7) FIG. 3C is a schematic top view of the metal mask in FIG. 3B.

DETAILED DESCRIPTION

- (8) Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.
- (9) In the following description, in order to clearly present the technical features of the present disclosure, the dimensions (such as length, width, thickness, and depth) of elements (such as layers, films, substrates, and areas) in the drawings will be enlarged in unusual proportions. Accordingly, the description and explanation of the following embodiments are not limited to the sizes and shapes of the elements presented in the drawings, but should cover the sizes, shapes, and deviations of the two due to actual manufacturing processes and/or tolerances. For example, the flat surface shown in the drawings may have rough and/or non-linear characteristics, and the acute angle shown in the drawings may be round. Therefore, the elements presented in the drawings in this case which are mainly for illustration are intended neither to accurately depict the actual shape of the elements nor to limit the scope of patent applications in this case.
- (10) Moreover, the words, such as “about”, “approximately”, or “substantially”, appearing in the present disclosure not only cover the clearly stated values and ranges, but also include permissible deviation ranges as understood by those with ordinary knowledge in the technical field of the invention. The permissible deviation range can be caused by the error generated during the measurement, where the error is caused by such as the limitation of the measurement system or the process conditions. In addition, “about” may be expressed within one or more standard deviations of the values, such as within $\pm 30\%$, $\pm 20\%$, $\pm 10\%$, or $\pm 5\%$. The word “about”, “approximately” or “substantially” appearing in this text can choose an acceptable deviation range or a standard deviation according to optical properties, etching properties, mechanical properties or other properties, not just one standard deviation to apply all the optical properties, etching properties, mechanical properties and other properties.
- (11) FIG. 1A is a schematic view of a calendering process employed by a method of manufacturing a metal mask according to at least one embodiment of this disclosure. Referring to FIG. 1A, in the method of manufacturing the metal mask according to this embodiment, first, a metal material **10** is calendered, so as to form a metal mask substrate **101**. Specifically, the metal material **10** can be fed to a calendering machine **200** for performing calendering, in which the calendering machine **200**

includes a plurality of calendering rollers **210** and a plurality of transferring rolls **290**.

(12) The metal material **10** can be a flexible metal strip or a rigid metal plate, in which some transferring rolls **290** can pass the metal material **10** between the calendering rollers **210**, so that two adjacent rotating calendering rollers **210** can calender the metal material **10** to form a metal mask substrate **101**. The other transferring rolls **290** can transfer the metal mask substrate **101** that is calendered, so that the metal mask substrate **101** leaves the calendering rollers **210** and is output from the calendering machine **200**.

(13) It is worth mentioning that the calendering machine **200** shown in FIG. **1A** can include four calendering rollers **210**, in which the four calendering rollers **210** can calender the same metal material **10** in the meanwhile. In addition, in this embodiment, the sizes of at least two calendering rollers **210** can be different. Taking FIG. **1A** for example, the sizes of two calendering rollers **210** on the left are significantly larger than the sizes of two calendering rollers **210** on the right. However, in other embodiment, the sizes of the calendering rollers **210** can be substantially the same, and thus, FIG. **1A** does not limit the sizes of the calendering rollers **210**.

(14) FIG. **1B** is a schematic partial enlarged view of the surface of the calendering roller in FIG. **1A**. Referring to FIGS. **1A** and **1B**, each of the calendering rollers **210** can include a calendering layer **211** and a roller **212**. In the same calendering roller **210**, the calendering layer **211** covers the outer surface of the roller **212**, so that the calendering layer **211** wraps the roller **212**, in which the calendering layer **211** can be attached to the roller **212** by adhesive, and the calendering layer **211** can be made of titanium.

(15) There may be a plurality of bumps **211p** which can press the metal material **10** formed on the surface of the calendering layer **211**, so as to form a plurality of grooves in the surface of the metal material **10**. The bumps **211p** have various sizes, and the width **211r** of the bump **211p** can range between 7 μm and 23 μm substantially. Moreover, in this embodiment, the bumps **211p** can be arranged irregularly, as shown in FIG. **1B**. However, in other embodiment, the bumps **211p** also can be arranged regularly, such as in a matrix.

(16) It is necessary to note that the bumps **211p** are distributed densely, in fact. However, FIG. **1B** is the schematic partial enlarged view of the surface of the calendering roller **210**, and FIG. **1B** is the simplified drawing of the surface of the calendering roller **210**. For example, the quantity of the bumps **211p** is reduced, so that FIG. **1B** can show the bumps **211p** simply and clearly, thereby facilitating marking the bump **211p** and the width **211r**. Therefore, FIG. **1B** is merely for illustration and not intended to accurately depict the actual appearance of the surface of the calendering roller **210**.

(17) FIG. **2A** is a schematic top view of the metal mask substrate formed via the calendering process in FIG. **1A**. Referring to FIG. **2A**, the metal mask substrate **101** includes a surface **110** and a plurality of grooves **130** formed in the surface **110**, where the grooves **130** are formed by the calendering rollers **210** pressing the metal material **10**. Thus, the grooves **130** all extend in the same direction. Taking FIG. **2A** for example, the grooves **130** can be distributed side by side and all extend in a first direction **D1**.

(18) The metal mask substrate **101** further includes a pair of long sides **191** opposite to each other and a pair of short sides **192** opposite to each other, in which the long sides **191** are connected to the short sides **192**. In this embodiment, the extending direction of the long sides **191** can be substantially parallel to the first direction **D1**, whereas the extending direction of the short sides **192** can be substantially parallel to a second direction **D2**, where the first direction **D1** can be substantially perpendicular to the second direction **D2**.

(19) It is necessary to note that as seen from the embodiment shown in FIG. **2A**, each of the grooves **130** has the length basically equal to the length of the long side **191**. However, in other embodiment, the length of at least one of the grooves **130** can be shorter than the length of the long side **191**. For example, during the above calendering process, the multiple bumps **211p** can stamp at least two collinear grooves **130** which are not connected. Accordingly, the length of one of the

grooves **130** shown in FIG. 2A can be shorter than the length of the long side **191** and may be not equal to the length of the long sides **191**.

(20) In addition, since the bumps **211p** in the same calendering roller **210** are actually distributed densely, the side-by-side grooves **130** are also distributed densely. However, FIG. 2A is the simplified drawing of the grooves **130**. For example, the quantity of the grooves **130** is reduced, so as to show the grooves **130** simply and clearly. Hence, FIG. 2A is merely for illustration and not intended to accurately depict both the actual appearance and the real distribution of the grooves **130**.

(21) Then, at least one sampling region **150** is selected from the surface **110** of the metal mask substrate **101**, so that the surface **110** has at least one sampling region **150**. The sampling region **150** can be marked on the surface **110**, i.e., the sampling region **150** is visible. Alternatively, the sampling region **150** can be not marked on the surface **110**, i.e., the sampling region **150** is invisible.

(22) For example, the sampling region **150** can be selected by an instrument, where the instrument can be an optical measurement instrument, such as optical microscope or optical dimension measuring device. The optical measurement instrument can directly select at least one sampling region **150** from the surface **110** of the metal mask substrate **101** without any mark, so the sampling region **150** cannot be seen by human eye.

(23) In the embodiment shown in FIG. 2A, a plurality of sampling regions **150** are selected from the surface **110** of the metal mask substrate **101**, so that the surface **110** has a plurality of sampling regions **150**, in which the sampling regions **150** can be distributed along the long sides **191**, i.e., the sampling regions **150** can be distributed in the first direction D1. However, in other embodiment, only one sampling region **150** is selected from the surface **110**, i.e., the surface **110** has only one sampling region **150**. Hence, the quantity of the sampling region **150** which the surface **110** has can be only one and not limited by FIG. 2A.

(24) FIG. 2B is a schematic top view within one of the sampling regions of the metal mask substrate in FIG. 2A. Referring to FIGS. 2A and 2B, the multiple grooves **130**, i.e., at least two grooves **130**, can be distributed in each of the sampling regions **150**, so there are at least two side-by-side and noncollinear grooves **130** formed in one of the sampling regions **150**, as shown in FIG. 0.2B.

(25) In this embodiment, the sampling region **150** can take the shape of a square substantially, in which the side **151** of the sampling region **150** may be about 5 mm, for example. In other words, the sampling region **150** can take the shape of a 5 mm×5 mm square substantially. The grooves **130** can be formed by the bumps **211p** pressing the metal material **10**, so that the width **131** of each of the grooves **130** is basically equal to the width **211r** of the bump **211p** (referring to FIG. 1B). Thus, the width **131** of each of the grooves **130** can range between 7 μm and 23 μm.

(26) It is necessary to note that the width **131** of each of the grooves **130** can range between 7 μm and 23 μm, and the side **151** of the sampling region **150** can be about 5 mm, so that FIGS. 2A and 2B apparently do not depict the sampling regions **150** and the grooves **130** on the same scale according to real size. Furthermore, the grooves **130** shown in FIG. 2B are also not drawn to scale based on enlarging FIG. 2A. Hence, FIGS. 2A and 2B are not intended to accurately depict the grooves **130** and the sampling regions **150**, and both the grooves **130** and the sampling regions **150** shown in FIGS. 2A and 2B are merely for illustration to help to describe the grooves **130** and the sampling regions **150**.

(27) In addition, since the bumps **211p** of the calendering roller **210** are distributed densely in fact, so that one of the grooves **130** can be formed by the multiple bumps **211p** calendering the metal material **10**. Hence, the shape of one single groove **130** is a bar, but the shape may not be a rectangle. For example, one single groove **130** can be formed by a plurality of circular and/or elliptical holes which are arranged in a line and connected, so that the groove **130** can have an undulate edge and a significantly inconstant width. Thus, the grooves **130** shown in FIG. 2B are

merely for illustration to help to describe structural features of this embodiment and not intended to accurately depict the grooves **130**.

(28) After one or more sampling regions **150** are selected, an average area ratio of the area of the grooves **130** within each sampling region **150** to the area of the sampling region is measured and obtained. During measuring and obtaining the average area ratio, first, in the same sampling region **150**, the area of the grooves **130** within the sampling region **150** and the area of the sampling region **150** can be measured and obtained, where the area of the grooves **130** is equal to the sum of areas of the grooves **130** within the same sampling region **150**.

(29) Taking FIG. 2B for example, first, the area of the sampling region **150** shown in FIG. 2B (i.e., side **151** × side **151**) and the area of all of the grooves **130** within the sampling region **150** can be measured. Then, an area ratio defined by the area of all of the grooves **130** within the sampling region **150** and the area of the sampling region **150** is calculated. For example, in FIG. 2B, the area of the sampling region **150** is A, and the area of all of the grooves **130** within the sampling region **150** is T, in which the area ratio is equal to T/A.

(30) According to the area ratio defined by the area of the grooves within the sampling region and the area of the sampling region, the average area ratio can be known. In detail, according to the definition of the area ratio, the area ratios of the sampling regions **150** are measured and obtained. Afterwards, an average of the area ratios of the sampling regions **150** is calculated, where the average is equal to the average area ratio and may be an arithmetic mean.

(31) Afterwards, the average area ratio is determined whether it is within an eligible range, so as to determine the metal mask substrate **101** can be ready for subsequent process, in which the eligible range is between 45% and 68%, for example. When the average area ratio is within the eligible range, e.g., between 45% and 68%, it means that the surface **110** of the metal mask substrate **101** has suitable roughness, thereby facilitating the subsequent process, such as photolithography.

(32) FIGS. 3A and 3B are schematic cross-sectional views of the metal mask substrate in FIG. 2A during photolithography. Referring to FIG. 3A, first, an initial photoresist layer **30** can be formed on the surface **110** of the metal mask substrate **101** during the photolithography. Since the surface **110** has the suitable roughness, the initial photoresist layer **30** covering the surface **110** can be firmly attached to the surface **110**.

(33) Referring to FIGS. 1A and 3A, in particular, in this embodiment, the metal material **10** can be calendered by two adjacent and rotating calendering rollers **210**, where each of the calendering rollers **210** has a plurality of bumps **211p**, so two opposite surfaces (including the surface **110**) of the metal mask substrate **101** have a plurality of grooves **130** apiece. Taking FIG. 3A for example, the metal mask substrate **101** further has a surface **111** opposite to the surface **110**, in which a plurality of (e.g., at least two) grooves **130** are also formed in the surface **111**.

(34) Like the surface **110**, the surface **111** also has at least one sampling region, where both the sampling region **150** and the sampling region of the surface **111** can have the same sizes and shapes. Moreover, the method of measuring and obtaining the average area ratio from the surface **111** is also the same as the method of measuring and obtaining that from the surface **110**, and thus not described again.

(35) After the average area ratio is obtained from the surface **111**, the average area ratio (obtained from the surface **111**) is determined whether it is within an eligible range, which may also be between 45% and 68%. When each of the average area ratios of both the surfaces **110** and **111** ranges between 45% and 68%, it means that both the surfaces **110** and **111** have suitable roughness, so that two initial photoresist layers **30** can cover and be attached to the surfaces **110** and **111** firmly.

(36) Referring to FIG. 3B, afterwards, the initial photoresist layers **30** are developed, so as to form two photoresist patterns **31**, in which the photoresist patterns **31** have a plurality of holes **31a** and **31b** (where FIG. 3B shows only one hole **31a** and one hole **31b**). In the embodiment shown in FIG. 3B, the upper photoresist pattern **31** has the plurality of holes **31a** exposing the surface **110**, and the

lower photoresist pattern **31** has the plurality of holes **31b** exposing the surface **111**, wherein the size of the hole **31a** is apparently smaller than the size of the hole **31b**.

(37) After the photoresist patterns **31** are formed, by using the photoresist patterns **31** as masks, the metal mask substrate **101** is etched, so that a plurality of the openings **170** (FIG. 3B shows one opening **170**) are formed in the surfaces **110** and **111** of the metal mask substrate **101**, thereby forming a metal mask **100**, in which each of the openings **170** extends from the surface **110** to the surface **111**. In other words, the openings **170** are formed by passing through the metal mask substrate **101**.

(38) Since each of the surfaces **110** and **111** has the suitable roughness, not only the initial photoresist layers **30** can cover and be attached to the surfaces **110** and **111** firmly, but also the photoresist patterns **31** can cover and be attached to the surfaces **110** and **111** firmly. Therefore, during etching the metal mask substrate **101**, the photoresist patterns **31** can effectively block the etchant from entering the area except the holes **31a** and **31b** of the photoresist patterns **31**, so that the sizes and the shape of two ends of the opening **170** can be substantially the same as the sizes and the shapes of the holes **31a** and **31b**.

(39) FIG. 3C is a schematic top view of the metal mask in FIG. 3B. Referring to FIGS. 3B and 3C, afterwards, the photoresist patterns **31** are removed, so that both the surfaces **110** and **111** can appear. So far, a metal mask **100** including the surfaces **110** and **111**, the pair of the long sides **191**, the pair of the short sides **192**, the plurality of grooves **130** and the plurality of openings **17** is basically complete.

(40) Since the surfaces **110** and **111** are still covered by the photoresist patterns **31** during etching the metal mask substrate **101**, both the metal mask **100** and the metal mask substrate **101** include the grooves **130** which basically have the same width (e.g., the width **131** shown in FIG. 2B), while the average area ratios measured and obtained from the surfaces **110** and **111** of the metal mask **100** are also within the eligible range, such as between 45% and 68%.

(41) It is worth mentioning that in the embodiment shown in FIGS. 3A and 3B, the initial photoresist layers **30** are all developed to form the photoresist patterns **31**. However, one of the calendaring rollers **210** in FIG. 1A can have no bump **211p**, so that only one side of the metal material **10** is pressed by the bumps **211p**, whereas the other side of the metal material **10** is not pressed by any bump **211p**. Accordingly, the grooves **130** can be formed only in one side of the metal mask substrate **101**, for example, formed only in the surface **110**, not in the surface **111**.

(42) When the grooves **130** are only formed in the surface **110**, the initial photoresist layer **30** covering the surface **111** cannot be developed, so that the initial photoresist layer **30** can keep covering the surface **111** during etching the metal mask substrate **101**, thereby preventing the etchant in contact with the surface **111**. Thus, the hole **31b** shown in FIG. 3B can be omitted, and the openings **170** can be formed by etching only one side (e.g., surface **110**) of the metal mask substrate **101**.

(43) Consequently, when the average area ratio obtained based on the sampling region is within the eligible range (e.g., between 45% and 68%), the surface of the metal mask substrate has the suitable roughness, so that during photolithography, the photoresist pattern can cover and be attached to the surface of the metal mask substrate firmly. Therefore, the plurality of openings of the metal mask can be basically formed depending on the holes of the photoresist pattern completely, so as to reduce or avoid the ineligible openings, thereby causing the metal mask suitable for manufacturing display panel, especially manufacturing high resolution display.

(44) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(45) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the

disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A metal mask, comprising: a surface, consisting of a plurality of sampling regions, wherein each of the plurality of sampling regions has a same area; a pair of long sides, opposite to each other, wherein an extending direction of the long sides is parallel to a first direction; a pair of short sides, opposite to each other, wherein an extending direction of the short sides is parallel to a second direction, and the first direction is perpendicular to the second direction, wherein an area of the surface is defined by the long sides and the short sides; a plurality of grooves formed in the surface, and all extending in the first direction, wherein for each of the plurality sampling regions, an average area ratio of an area of the plurality of grooves within the sampling region to an area of the sampling region ranges between 45% and 68%, thereby increasing a suitable roughness of the surface configured to facilitate a subsequent process of the metal mask, wherein a width extending in the second direction of each of the plurality of grooves ranges between 7 μm and 23 μm ; and a plurality of openings, formed in the surface.
 2. The metal mask of claim 1, wherein the plurality of grooves are formed by pressing the metal mask with a calendering roller having a plurality of bumps.
 3. The metal mask of claim 1, wherein the subsequent process is photolithography.
-